

Section : Biology

Q.1 Which one of the following is known as cytokine barriers?

1. Skin of our body
2. Acid in our stomach
3. Leucocytes of our body
4. Interferons

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Arrange the steps of biopsy and histopathology of tissue involved in detection of cancer.

- (A) A thin section of tissue is stained
- (B) Freezing of tissue in liquid nitrogen
- (C) A piece of the suspected tissue cut into thin sections
- (D) Examination under microscope

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (C), (A), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Hepatitis B vaccine is produced from:

1. Yeast
2. Bacteria
3. Virus
4. Insect

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Match List-I with List-II

List-I	List-II
Aspect	Explanation
(A) Productivity	(I) Photosynthetically active radiation
(B) Nutrient Cycling	(II) Breakdown of organic matter
(C) Energy Flow	(III) Rate of biomass production
(D) Decomposition	(IV) Sedimentary

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Of the incident solar radiation, less than _____ of it is photosynthetically active radiation (PAR).

1. 20 percent
2. 70 percent
3. 50 percent
4. 30 percent

Options 1. 1

2. 2
3. 3
4. 4

Q.6 What is the role of fungal symbiont in mycorrhiza?

1. Absorbs phosphorus from the soil and passes it to the plant
2. Adds organic matter to the soil
3. Fixes atmospheric nitrogen
4. Increases soil fertility

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Match List-I with List-II

List-I

List-II

Habitat

Animal

(A) Sunderbans (I) Lion

(B) Periyar (II) Wild ass

(C) Rann of Kutch (III) Bengal tiger

(D) Gir (IV) Elephant

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The membrane surrounding the secondary oocyte is called-

1. Antrum
2. Granulosa layer
3. Theca
4. Zona pellucida

Options 1. 1

2. 2
3. 3
4. 4

Q.9 The age distribution plotted for the population is known as the 'Age pyramid'. Select the three types of age pyramids based on the shape and age groups:

- (A) Expanding
- (B) Inverted
- (C) Stable
- (D) Declining

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.10 Which of the following statements are correct about energy flow?
(A) Net primary productivity is available to consumers.
(B) Secondary productivity increases with each trophic level.
(C) The growth of the food chain is limited by the inefficiency of energy transfer.
(D) About 90% energy at one trophic level does not appear at the next level.
Choose the correct answer from the options given below:
1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.11 In the double-stranded DNA, percentage of adenine is 30%. Calculate the percentage of cytosine present in it?
1. 20%
2. 30%
3. 40%
4. 70%

- Options 1. 1
2. 2
3. 3
4. 4

Q.12 In *lac* operon, a structural gene is-
1. Monocistronic
2. Polycistronic
3. Initiator
4. Anticodon

- Options 1. 1
2. 2
3. 3
4. 4

Q.13 The Roquefort cheese are ripened by growing a specific _____ on them, which gives them a particular flavour.
1. Bacteria
2. Fungus
3. Protozoa
4. Virus

- Options 1. 1
2. 2
3. 3
4. 4

Q.14 The fluid filled cavity present in the secondary follicle is known as:
1. Zona pellucida
2. Antrum
3. Theca interna
4. Corpus luteum

- Options 1. 1
2. 2
3. 3
4. 4

Q.15 What are seen as "beads on string" structure when viewed under an electron microscope ?

1. Nucleoid
2. Chromatin
3. Base pairs
4. Nucleosomes

Options 1. 1

2. 2
3. 3
4. 4

Q.16 A combination of progestogen-estrogen is used in:

1. Condoms
2. Emergency contraceptives pills or IUDs
3. Spermicidal creams
4. Copper releasing IUDs

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Verhulst-Pearl Logistic Growth is described by the following equation:

$dN/dt = rN (K-N)/K$, where 'r' denotes the-

1. Carrying capacity
2. Intrinsic rate of natural increase
3. Population density
4. Mortality rate

Options 1. 1

2. 2
3. 3
4. 4

Q.18 The 'hilum' represents the junction between:

1. Ovule and funicle
2. Ovule and integument
3. Micropyle and chalaza
4. Nucellus and funicle

Options 1. 1

2. 2
3. 3
4. 4

Q.19 The chromosome number per cell is 46 in:

1. Primary oocyte
2. Secondary oocyte
3. Polar body
4. Ovum

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Which of the following are Mendelian disorders?

- (A) Cystic fibrosis
- (B) Phenylketonuria
- (C) Adenosine deaminase deficiency
- (D) Haemophilia

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.21 STIs are reported to be very high among persons in which age group?

1. 10 to 15 years
2. 15 to 24 years
3. 24 to 30 years
4. 30 to 35 years

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Which of the following statements are correct?

- (A) A lion eating a deer and a sparrow feeding on grains are ecologically similar consumers.
(B) Predator star fish *Pisaster* helps in maintaining the species diversity of some invertebrates.
(C) Predators ultimately lead to the extinction of prey species.
(D) Production of chemicals such as nicotine and strychnine by the plants are metabolic disorders.
Choose the correct answer from the options given below:

1. (B) and (D) only
2. (A) and (D) only
3. (C) and (D) only
4. (A) and (B) only

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Match List-I with List-II

List-I

Genetic disorder

(A) Thalassemia

(B) Sickle-cell anemia

(C) Klinefelter's syndrome

(D) Turner's syndrome

List-II

Cause

(I) 45+XO

(II) Formation of abnormal hemoglobin

(III) Substitution of glutamic acid by valine

(IV) 47, XXY

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Which disease is caused by filarial worm, that leads to chronic inflammation?

1. Ascariasis
2. Elephantiasis
3. Amoebiasis
4. Ringworm

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Identify the correct hormone releasing IUDs?

1. Multiload 375
2. LNG-20
3. Lippes loop
4. Cu7

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Which one is not correct about withdrawal syndrome?

1. Dependence is the tendency of the body to manifest a characteristic and unpleasant.
2. It is characterized by anxiety, shakiness, nausea and sweating.
3. The person suffers bouts of fever, diarrhoea and weight loss.
4. It can be severe and even life-threatening.

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Match List-I with List-II

List-I

List-II

Evolutionary stages of human species Time of existence/evolution

- | | |
|-------------------------------|----------------------------------|
| (A) <i>Dryopithecus</i> | (I) 100,000-40,000 years ago |
| (B) <i>Australopithecines</i> | (II) About 1.5 million years ago |
| (C) <i>Homo erectus</i> | (III) About 2 million years ago |
| (D) Neanderthal man | (IV) About 15 million years ago |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 "Toddy" is a traditional drink which is made by-

1. Fermenting bamboo-shoots
2. Fermenting sap from palm
3. Fermenting fishes
4. Fermenting cheese

Options 1. 1

2. 2
3. 3
4. 4

Q.29 In the species-area relationship equation, $\log S = \log C + Z \log A$, 'Z' denotes -

1. Species richness
2. Regression coefficient
3. Intercept
4. Area

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Arrange in increasing order of number of species to represent global biodiversity?

- (A) Molluscs

- (B) Crustaceans
(C) Insects
(D) Other animal groups

Choose the correct answer from the options given below:

1. (A), (C), (B), (D)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The sex chromosome of a female bird has

1. XZ
2. WW
3. ZW
4. ZZ

Options 1. 1

2. 2
3. 3
4. 4

Q.32 The primary oocytes are arrested at which stage of meiosis before birth?

1. Metaphase I
2. Prophase II
3. Prophase I
4. Anaphase I

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which of the followings were dominant during the 'Jurassic' period?

- (A) Ferns
- (B) Seed ferns
- (C) Conifers
- (D) Dinosaurs

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Aphids and mosquitoes can be controlled by, respectively:

1. Ladybirds and dragonflies
2. Ladybirds and butterflies
3. Catterpillars and birds
4. Birds and dragonflies

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Arrange the stages of embryo development in a dicot plant.

- (A) Heart shaped embryo
- (B) Globular embryo
- (C) Zygote
- (D) Mature embryo

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (C), (B), (D), (A)
3. (B), (A), (C), (D)
4. (C), (B), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Arrange the following geological periods in latest to oldest duration of existence.

- (A) Triassic
- (B) Carboniferous
- (C) Cretaceous
- (D) Permian

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (D), (C), (B)
3. (B), (A), (D), (C)
4. (C), (A), (D), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Identify the regions, where sacred groves are not found?

1. Khasi and Jaintia hills in Meghalaya
2. Western Ghat regions of Karnataka and Maharashtra
3. Caves of Ajanta and Elora
4. Sarguja, Chanda and Bastar areas of Madhya Pradesh

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Match List-I with List-II

List-I	List-II
RNA type	Role/function
(A) mRNA	(I) Initiates transcription
(B) tRNA	(II) Structural and catalytic role during translation
(C) rRNA	(III) Provides a template
(D) RNA Polymerase	(IV) Reads the genetic code

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Identify the correct sequence of formation of embryo sac or female gametophyte in flowering plants?

- (A) Female gametophyte
- (B) 2-nucleate embryo sac
- (C) Megaspore mother cell
- (D) 4-nucleate embryo sac

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (D), (C), (B), (A)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 Which enzyme modified by genetic engineering is used as a 'clot buster'?

- 1. Lipase
- 2. Pectinase
- 3. Streptokinase
- 4. Statins

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.41 Read the passage carefully and answer the questions based on the passage:

Pest Resistance Plants

Various nematodes parasitize a wide variety of plants and animals, including human beings. A nematode, *Meloidogyne incognita*, infects the roots of tobacco plants and causes a great reduction in yield. A novel strategy was adopted to prevent this infestation which was based on the process of RNA interference (RNAi). RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing). Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat). (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice. Genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem. Therefore, the Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes).

RNA interference involves:

- 1. Synthesis of cDNA and RNA using reverse transcriptase
- 2. Silencing of specific mRNA due to complementary RNA
- 3. Interference of RNA in synthesis of DNA
- 4. Synthesis of mRNA from DNA

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.42 Read the passage carefully and answer the questions based on the passage:

Pest Resistance Plants

Various nematodes parasitize a wide variety of plants and animals, including human beings. A nematode, *Meloidogyne incognita*, infects the

roots of tobacco plants and causes a great reduction in yield. A novel strategy was adopted to prevent this infestation which was based on the process of RNA interference (RNAi). RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing). Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat). (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice. Genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem. Therefore, the Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes).

Genetic modification does not include -

1. Increased efficiency of mineral usage by plants
2. Reduced reliance on chemical pesticides
3. Helped to increase the post harvest losses
4. Enhanced nutritional value of food

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the passage carefully and answer the questions based on the passage:

Pest Resistance Plants

Various nematodes parasitize a wide variety of plants and animals, including human beings. A nematode, *Meloidogyne incognita*, infects the roots of tobacco plants and causes a great reduction in yield. A novel strategy was adopted to prevent this infestation which was based on the process of RNA interference (RNAi). RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing). Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat). (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice. Genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem. Therefore, the Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes).

Which one infects the roots of tobacco plants and causes a great reduction in yield?

1. *Bacillus thuringiensis*
2. *Meloidogyne incognita*
3. Insecticidal protein
4. cry I Ac

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the passage carefully and answer the questions based on the passage:

Pest Resistance Plants

Various nematodes parasitize a wide variety of plants and animals, including human beings. A nematode, *Meloidogyne incognita*, infects the roots of tobacco plants and causes a great reduction in yield. A novel strategy was adopted to prevent this infestation which was based on the process of RNA interference (RNAi). RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing). Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat). (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice. Genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem. Therefore, the Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes).

Bacillus thuringiensis produces proteins that do not kill certain insects such as

1. Lepidopterans
2. Coleopterans
3. Hymenopterans
4. Dipterans

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the questions based on the passage:

Pest Resistance Plants

Various nematodes parasitize a wide variety of plants and animals, including human beings. A nematode, *Meloidogyne incognita*, infects the roots of tobacco plants and causes a great reduction in yield. A novel strategy was adopted to prevent this infestation which was based on the process of RNA interference (RNAi). RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing). Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat). (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice. Genetic modification of organisms can have unpredictable results when such organisms are introduced into the ecosystem. Therefore, the Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee), which will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes).

The Indian Government organization GEAC stands for -

1. Genetics Engineered Approval Committee
2. Genetic Engineering Approved Committee
3. Genetic Engineer Approval Committee
4. Genetic Engineering Approval Committee

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the given paragraph carefully and answer the following questions:
Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus). There are some basic steps in genetically modifying an organism (i) identification of DNA with desirable genes; (ii) introduction of the identified DNA into the host; (iii) maintenance of introduced DNA in the host and transfer of the DNA to its progeny. *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol. Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure because it requires simultaneous plating on two plates having different antibiotics. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. In this, a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme, which is referred to as insertional inactivation.

Agrobacterium tumefaciens, a pathogen of several dicot plants, is able to deliver a piece of DNA known as:

1. R-DNA
2. C-DNA
3. T-DNA
4. ds-DNA

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the given paragraph carefully and answer the following questions:
Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus). There are some basic steps in genetically modifying an organism (i) identification of DNA with desirable genes; (ii) introduction of the identified DNA into the host; (iii) maintenance of introduced DNA in the host and transfer of the DNA to its progeny. *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol. Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure because it requires simultaneous plating on two plates having different antibiotics. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. In this, a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme, which is referred to as insertional inactivation.

Which one of the following steps is not involved in genetical modification of an organisms?

1. Transfer of the DNA to its progeny
2. Identification of DNA
3. Maintenance of introduced DNA in the host
4. Introduction of the identified DNA into the host

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the given paragraph carefully and answer the following questions:
Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus). There are some basic steps in genetically modifying an organism (i) identification of DNA with desirable genes; (ii) introduction of the identified DNA into the host; (iii) maintenance of introduced DNA in the host and transfer of the DNA to its progeny. *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol. Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure because it requires simultaneous plating on two plates having different antibiotics. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. In this, a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme, which is referred to as insertional inactivation.

Which enzyme is not used to lyse the cells for extraction of DNA?

1. Lysozyme
2. Cellulase
3. Protease
4. Chitinase

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the given paragraph carefully and answer the following questions:
Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus). There are some basic steps in genetically modifying an organism (i) identification of DNA with desirable genes; (ii) introduction of the identified DNA into the host; (iii) maintenance of introduced DNA in the host and transfer of the DNA to its progeny. *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol. Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure because it requires simultaneous plating on two plates having different antibiotics. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. In this, a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme, which is referred to as insertional inactivation.

Which one gives blue colored colonies if the plasmid in the bacteria does not have an insert?

1. The presence of cancerous cells
2. The presence of agarose
3. The presence of chromogenic substrate
4. The presence of ribonuclease

Options 1. 1

2. 2

3. 3

Q.50 Read the given paragraph carefully and answer the following questions:
Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus). There are some basic steps in genetically modifying an organism (i) identification of DNA with desirable genes; (ii) introduction of the identified DNA into the host; (iii) maintenance of introduced DNA in the host and transfer of the DNA to its progeny. *Agrobacterium tumefaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol. Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure because it requires simultaneous plating on two plates having different antibiotics. Therefore, alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate. In this, a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme, which is referred to as insertional inactivation.

The purified DNA ultimately precipitates after the addition of:

1. Methanol
2. Chilled ethanol
3. Protease
4. Ethidium bromide

Options 1. 1

2. 2

3. 3

4. 4